# VEER ARORA

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## PROFILE

Data and backend engineer, B.Tech Computer Science & Design (2026), who builds systems that check their own output. Built a daily NSE and BSE price pipeline on DuckDB, dbt and GitHub Actions whose loads are idempotent and restatable, an API connector that resumes safely after a crash, a bank-statement ledger that proves it balances to the paisa before it categorises anything, and a RAG system at recall@5 0.77 against a measured ceiling of 0.85 on 60 questions. At L&T Finance, validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow, ahead of its production release.

## EDUCATION

**Madhav Institute of Technology & Science, Gwalior**, B.Tech, Computer Science & Design, 2022 – 2026

Shanti Niketan HS School, Vidisha, Class 12 (2021) • Trinity Convent Sr. Sec. School, Vidisha, Class 10 (2019)

## SKILLS

**Languages & Data** Python, SQL including recursive CTEs, DuckDB, dbt, SCD type 2 dimensions, Dagster, Pandas, NumPy, data cleaning and validation, EDA

**AI & LLM** LLM tool-calling agents, RAG, LangGraph, evaluation harnesses (tool-selection, task-success, grounding, cost and latency), held-out test splits, OpenAI, Anthropic, Google Gemini and Groq APIs

**Backend & Web** FastAPI, PostgreSQL, row-level security, SQLAlchemy, Alembic, REST APIs, GitHub Actions CI, pytest, Docker, Git, React, TypeScript

**Documents & Extraction** PDF table recovery with pdfplumber (ruled and borderless), encrypted PDFs, scanned-vs-digital triage and OCR, ReportLab PDF generation, openpyxl

**Cloud & MLOps** Google Cloud Platform, Kubeflow, pipeline testing, UAT

**Analytics & ML** correlation analysis, descriptive statistics, outlier and anomaly detection, scikit-learn

**Reporting & Tools** Power BI, Matplotlib, Seaborn, Streamlit, SQLite, n8n, ffmpeg, OpenCV

## EXPERIENCE

### L&T Finance, Data Analyst Intern **Jun 2025 – Aug 2025**

- Validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow spanning EWS, Banking, Self-Cure and Collections workflows, ahead of its production release.
- Caught an output-formatting defect in the EWS model that left its output column blank downstream, a failure that surfaces as missing data rather than as an error, and drove root-cause analysis with the developers and MLOps engineers who owned the pipeline.
- Wrote SQL verification across pipeline stages: data-integrity and row-level checks that separated “the pipeline ran” from “the pipeline produced the right numbers”.
- Ran end-to-end pipeline validation and UAT in Python and SQL, covering test planning, defect tracking and deployment-readiness review.

### E-Cell, IIT Bombay, Campus Executive **Aug 2023 – Feb 2024**

- Promoted entrepreneurship programs, workshops and competitions, and coordinated outreach, event logistics and mentor coordination across the student community.

## PROJECTS

### bhavcopy-pipeline: a daily market-data pipeline **github.com/veer0608/bhavcopy-pipeline**

*Python · DuckDB · dbt · pytest · GitHub Actions*

- Built a daily pipeline for NSE and BSE end-of-day prices with no model in it: each file is landed as an immutable gzip, loaded into DuckDB, then modelled and tested with dbt, on a GitHub Actions schedule. 659,936 rows across 77 trading days per exchange, to 19 October 2026.
- Made the load **idempotent and restatable**, and tested both: re-running a day changes nothing, an incremental build equals a full refresh, and a republished file replaces its day whole. 33 dbt data tests.
- Found that the dashboard's top losers often listed stocks that had not fallen: on a split or bonus day the exchange file reports the previous close in old shares, so a 2:1 bonus read **-65.90%**. Added NSE corporate actions and a split-adjusted daily change in dbt; split stocks in the top-10 losers went from **30 of 144 exchange-days to 1**, a genuine fall. A uniqueness test caught duplicated rows from the BSE-to-NSE ISIN match before it shipped.

### syncline: an incremental API connector **github.com/veer0608/syncline**

*Python · DuckDB · Dagster · REST APIs · pytest*

- Built a connector that syncs GitHub issues and pull requests into DuckDB incrementally. Each page and its checkpoint commit in **one transaction**, so a run killed anywhere resumes with nothing lost or doubled. Rate limits are waited out, server errors back off with jitter, and new fields are recorded rather than fatal. Scheduled as a Dagster asset; 18 offline tests against a fault-injecting fake API.
- The first live run reported success but had **silently skipped 7,553 rows**: GitHub's updated_at does not always follow its own sort order, and one out-of-place row made the cursor jump 18 months. Found it by comparing month-by-month counts with GitHub's own; after the fix a full sync holds **22,425 rows, exactly GitHub's count**, and a hard kill plus resume matches a clean run.

### moneytrail: a bank-statement ledger that provably balances **github.com/veer0608/moneytrail**

*Python · pdfplumber · SQLite · pytest · GitHub Actions*

- Built a local-first tool that turns bank statements into a ledger. It starts with a **reconciliation gate** rather than a categoriser: every parsed row is checked against arithmetic the bank already published, so a dropped row surfaces as a line number and a rupee delta instead of a quietly wrong total. Money is an integer count of paise throughout, with no floats in the codebase.
- Parses CSV, spreadsheets and real bank PDFs, including password-protected files and borderless tables, and rejects scanned PDFs rather than guessing. Verified on a real bank export: 225 transactions across three months reconcile to the paisa, and first contact with real data surfaced four defects no synthetic fixture could have found. Each is now a test.
- 529 tests, CI green on Python 3.11 and 3.12. Also shipped as a paid n8n template where the model only assigns column roles and arithmetic checks every row; in a live run it caught a salary row the model had mislabelled.

**AgentOps: an AI support engineer that takes gated actions** **github.com/veer0608/agentops**

*Python · FastAPI · LangGraph · SQLAlchemy · Alembic*

- Built an AI support agent that takes real actions (create and update tickets, escalate) behind a policy gate. Each action is auto-executed, routed to a human review queue, or blocked, based on business rules and confidence thresholds. Writes are idempotent and recorded in an append-only audit log.
- Made the model provider-agnostic, with OpenAI, Anthropic and Google Gemini swappable through one config value, and implemented a second agent on LangGraph behind the same interface. The eval harness shows both implementations score identically.
- Wrote a scenario-based evaluation harness scoring six dimensions (tool-selection precision, recall and F1, task-success, escalation accuracy, citation-grounding, cost, latency) and persisted every run for before-and-after tracking. FastAPI, SQLAlchemy and Alembic on SQLite in development and Postgres in production; 38 automated tests.
- Ran the suite eight times against a live model rather than once. Tool-selection F1 rose from a 0.60 baseline to 0.78 on the mean, and citation-grounding held at 1.000 with zero deviation across 80 scenario-runs. Run-to-run spread was 0.11, wider than the effect being measured, so the writeup reports a mean and a spread rather than whichever single pass looked best.

**CiteRAG: retrieval-augmented QA with a golden-set eval** **github.com/veer0608/citerag**

*Python · FastAPI · SQLite-vec · bge embeddings and cross-encoder*

- Built a retrieval-augmented QA system over messy 10-K PDFs, where every answer is grounded in cited source chunks with page numbers.
- Created a golden-set eval harness (recall@k, MRR). On 30 questions, chunking, a cross-encoder re-ranker and hybrid keyword search raised recall@5 from 0.37 to 0.73; on a harder 60 it reaches 0.77 against a measured ceiling of 0.85. On the 60, the largest gain came from fixing PDF text extraction.
- Local embeddings and re-ranker on SQLite-vec, so the retrieval path and the whole eval harness run without a paid API. 80 tests, CI on every push.

**schemablind: a schema-blind SQL agent** **github.com/veer0608/schemablind**

*Python · SQLite · BIRD benchmark · pytest*

- Built a SQL agent that is given no schema. It explores a database it has never seen through four read-only verbs (list tables, describe, sample rows, run query) and composes its own answer, recovering from its own failed queries as it goes.
- Scored on BIRD execution accuracy: the agent's rows against the reference query's rows on the same database, with no judge model deciding whether the SQL looks right. 40% of questions are held back as a test split assigned by hash of the question text, so a prompt tuned on failures cannot choose what it is marked on. First full run: 64.7% on the 266-question dev half with Gemini 3.5 Flash-Lite, about half a cent per question; the held-out half is not yet scored.
- Tested the harness before trusting it: an oracle solver must score 100% and a mute solver 0%, asserted in CI on every push. Model-composed SQL runs behind four independent sandbox layers, and CI asserts the database is byte-identical after attempted writes. A run that exhausts its token budget is reported as unscored rather than as a partial number. 203 tests.

**vidsmith: script to narrated, captioned video** **github.com/veer0608/vidsmith**

*Python · edge-tts · ffmpeg · Gemini vision*

- Built a pipeline that turns a markdown script into a delivery-ready mp4: neural narration, a shot per scene, burned-in word-timed captions, music ducked under the voice, plus an .srt file and a draft title, description and chapters.
- Caption and cut timings come from the TTS engine's own word-boundary events rather than from transcribing the audio it had just produced, so there is no Whisper dependency and nothing to drift. Shots cut on the sentence boundaries those same events report, and candidate stock clips are re-ranked by a vision model against the narration line, because stock search ranks by popularity rather than by what is in frame.

## ALSO ON GITHUB

**finquery-eval-lab**, schema-constrained text-to-query for financial questions. The model only translates language into a validated query; deterministic code does the arithmetic over integer paise and returns the evidence rows. Explicit refusals, model-versus-baseline evals, cost and latency tracked. · **geojit-report-app**, turns a financial context document into an auto-filled equity-research PDF (FastAPI and ReportLab), with rule-based extraction as a fallback so generation never fails. · **data-analysis-agent**, automated EDA behind an API: profile, server-rendered charts, IsolationForest and Z-score outliers, downloadable PDF report. · **AgencyDesk**, multi-tenant SaaS where tenant isolation is enforced by PostgreSQL row-level security rather than by application code. 70 tests prove cross-tenant access is impossible.

## CERTIFICATIONS & ACTIVITIES

Affective Computing, NPTEL (Jan–Apr 2024) • Digital Design with Verilog, NPTEL (Jan–Apr 2025) • Rational Thinking, Udemy

R&D Coordinator, Startup Cell, MITS (Jan 2023 – Jun 2026) • Core Team Member, Digital Learning Group, MITS (Feb 2023 – Jul 2024) • State-level swimmer

Languages: English, Hindi